

Pipedrive Integration

See also: [TESTING_GUIDE.md](#)

1. Overview

1.1 Description

Pipedrive Integration connects **Pipedrive CRM** with **Newo Platform** and lets a Newo agent manage appointment-style activities inside Pipedrive.

It currently supports:

- checking availability by reading existing Pipedrive activities
- booking appointments with automatic contact lookup or contact creation
- cancelling booked appointments
- identifying existing callers by phone and loading future bookings into persona memory
- token-based auth and connector-based OAuth auth

The integration is aimed at **service businesses and sales teams** that want a conversational agent to schedule, confirm, and cancel meetings in Pipedrive.

1.2 Use Cases

- When a caller asks for available times -> the agent reads Pipedrive activities and computes free slots
- When a caller confirms a slot -> the agent creates or reuses a Pipedrive person and books an activity
- When a caller wants to cancel -> the agent deletes the matching Pipedrive activity
- When the caller phone number is known -> the agent loads the matching person and future bookings into prompt context
- When the session starts and preload is enabled -> the agent silently preloads nearest available slots

1.3 Supported Features

Feature	Supported	Notes
Availability check	✓	Uses <code>activities_list</code> and computes free slots locally
Booking creation	✓	Uses <code>persons</code> + <code>activities</code> APIs
Booking cancellation	✓	Deletes activity by <code>booking_id</code>
Existing client detection	✓	Phone-based lookup through <code>persons/search</code>
Prompt context preload	✓	Future bookings injected into prompt sections
Session-start slot preload	✓	Controlled by <code>pipedrive_check_availability_on_conversation_start</code>
Token auth	✓	<code>x-api-token</code> header through <code>http</code> connector

OAuth auth	✓	oauth_code + create_extension_connector through api connector
Reschedule flow	✗	Use cancel + rebook
Deal / pipeline management	✗	Not implemented
Multi-location routing	✗	Single Pipedrive workspace/account per integration

2. Requirements

Before setup:

- active Pipedrive account
- for Token auth: a Pipedrive API token
- for OAuth auth: a Pipedrive app authorization flow that yields oauth_code
- a correct tenant domain in pipedrive_base_api_url , because runtime requests target the workspace-specific Pipedrive URL even in OAuth mode

3. How to Setup

3.1 Authentication Modes

The integration has two auth modes:

1. Token

- operator sets pipedrive_auth_type = Token
- runtime sends x-api-token
- this is the simplest direct API path

2. OAuth

- operator sets pipedrive_auth_type = OAuth
- operator pastes pipedrive_oauth_code
- SetupFlow/_setupConnectorSkill creates a Pipedrive extension connector via create_extension_connector
- the visible authorization link currently uses placeholder app credentials in the repo and should be replaced with the real Pipedrive app credentials when they are known
- runtime API calls then go through the api connector instead of the raw token header path

3.2 Operator-Facing Attributes

Attribute	Required	Description
pipedrive_auth_type	Yes	OAuth or Token
pipedrive_api_token	Required for Token	Pipedrive API token used in x-api-token header
pipedrive_oauth_code	Required for OAuth	One-time authorization code used to create the OAuth connector

pipedrive_base_api_url	Yes	Required workspace URL such as https://your-company.pipedrive.com
project_business_time_zone	Yes	Shared project timezone used for slot interpretation and UTC conversion before booking
pipedrive_show_for_days	No	Availability window size, default 1
pipedrive_duration	No	Default appointment duration in minutes, default 30
pipedrive_enable_availability	No	Enable / disable availability tool
pipedrive_enable_booking	No	Enable / disable booking tool
pipedrive_enable_cancellation	No	Enable / disable cancellation tool
pipedrive_check_availability_on_conversation_start	No	Silent slot preload at session start
pipedrive_setup_scenarios	No	Publish default canvas/library content

3.3 System-Managed Hidden Attributes

These are not meant for normal operator editing but are part of the integration contract:

Attribute	Purpose
pipedrive_setup_persona_id	Setup persona used for setup-time operations
pipedrive_availability_extraction_schema	LLM extraction schema for availability
pipedrive_availability_extraction_instructions	LLM extraction instructions for availability
pipedrive_booking_extraction_schema	LLM extraction schema for booking
pipedrive_booking_extraction_instructions	LLM extraction instructions for booking
pipedrive_cancellation_extraction_schema	LLM extraction schema for cancellation
pipedrive_cancellation_extraction_instructions	LLM extraction instructions for cancellation

3.4 What SetupFlow Does

On publish, `SetupFlow` :

- registers both `api` and `http` connectors named `pipedrive_connector`
- creates/maintains the hidden setup persona
- publishes operator-facing and hidden attributes
- registers custom Newo tools through `_setupToolsSkill`

- when `pipedrive_auth_type = OAuth` , runs `_setupConnectorSkill` to create the OAuth extension connector

4. How to Use

4.1 Triggers

Trigger	Event	Purpose
Setup	<code>project_publish_finish</code>	Initialize attributes, tools, connectors
Setup callback	<code>result_success</code> on <code>api/pipedrive_connector</code>	Mark OAuth connector as ready
Availability	<code>pipedrive_check_availability</code>	Run availability lookup
Booking	<code>pipedrive_book_appointment</code>	Create a booking
Cancellation	<code>pipedrive_cancel_appointment</code>	Cancel a booking
Existing caller lookup	<code>define_user_phone_number</code>	Find person by phone and preload bookings
Session preload	<code>session_started</code>	Optional silent availability preload
Metadata	<code>get_project_metadata</code>	Return integration version

4.2 External Actions / API Calls

Action	Method	Path	Notes
Search person by phone	GET	<code>/api/v2/persons/search</code>	Used in existing-client and booking lookup paths
Create person	POST	<code>/api/v2/persons</code>	Uses <code>emails</code> / <code>phones</code> in v2 shape
List activities	GET	<code>/api/v2/activities</code>	Availability and existing-booking reads
Create activity	POST	<code>/api/v2/activities</code>	Booking create path
Delete activity	DELETE	<code>/api/v2/activities/{id}</code>	Cancellation path

4.3 Important Runtime Rules

- The integration uses **custom tools**, not generic NAF booking tools.
- `search_booking_tool` is still expected to stay available in ConvoAgent for booking lookup.
- Availability and booking extractions are LLM-assisted through hidden JSON schemas and extraction prompts.
- Booking now converts the caller-confirmed local time into **UTC** before sending `due_date` and `due_time` to Pipedrive.
- Activity creation in `v2` uses `participants` with a primary participant instead of writable `person_id` .

4.4 Architecture Diagrams

Overall Sequence

```
sequenceDiagram
    participant U as User
    participant C as ConvoAgent
    participant P as PipedriveAgent
    participant API as Pipedrive API

    Note over P: Setup
    C->>P: project_publish_finish
    P->>P: _initSkill + _setupToolsSkill
    alt OAuth mode
        P->>API: create_extension_connector
        API-->>P: result_success
        P->>P: mark oauth_ready
    end
    end

    Note over U,API: Existing caller lookup
    C->>P: define_user_phone_number
    P->>API: GET /api/v2/persons/search
    API-->>P: person data
    P->>API: GET /api/v2/activities
    API-->>P: future activities
    P->>C: prompt sections updated

    Note over U,API: Availability
    U->>C: asks for time slots
    C->>P: pipedrive_check_availability
    P->>API: GET /api/v2/activities
    API-->>P: booked activities
    P->>P: compute free slots
    P->>C: urgent_message with exact slots

    Note over U,API: Booking
    U->>C: confirms slot
    C->>P: pipedrive_book_appointment
    P->>API: GET /api/v2/persons/search or POST /api/v2/persons
    API-->>P: person id
    P->>P: convert local booking time to UTC
    P->>API: POST /api/v2/activities
    API-->>P: activity id
    P->>C: urgent_message booking success

    Note over U,API: Cancellation
    U->>C: asks to cancel
    C->>P: pipedrive_cancel_appointment
    P->>API: DELETE /api/v2/activities/{id}
    API-->>P: delete success
    P->>C: urgent_message cancellation success
```

AvailabilityFlow

```
flowchart LR
    E["pipedrive_check_availability or session_started"] --> G{"feature enabled?"}
    G -->|"No"| STOP["urgent_message / Return"]
    G -->|"Yes"| X["extract requested date/time"]
    X --> API["GET /api/v2/activities"]
    API --> OK{"success?"}
    OK -->|"No"| ERR["ResultErrorSkill"]
    OK -->|"Yes"| CALC["compute free slots from busy activities"]
    CALC --> OUT["ResultSuccessSkill updates prompt + urgent_message"]
```

BookingFlow

```
flowchart LR
    E["pipedrive_book_appointment"] --> G{"feature enabled?"}
    G -->|"No"| STOP["urgent_message / Return"]
    G -->|"Yes"| X["extract bookingDateTime + customerInfo"]
    X --> F{"person already known?"}
    F -->|"No"| C["POST /api/v2/persons"]
    F -->|"Yes"| U["reuse person_id"]
    C --> U
    U --> T["convert local time to UTC"]
    T --> A["POST /api/v2/activities with participants"]
    A --> OK{"success?"}
    OK -->|"No"| ERR["ResultErrorSkill"]
    OK -->|"Yes"| OUT["ResultSuccessSkill stores booking + urgent_message"]
```

CancellationFlow

```
flowchart LR
    E["pipedrive_cancel_appointment"] --> X["extract booking id / datetime"]
    X --> B{"matching booking found?"}
    B -->|"No"| ERR["ResultErrorSkill"]
    B -->|"Yes"| API["DELETE /api/v2/activities/{id}"]
    API --> OK{"success?"}
    OK -->|"No"| ERR
    OK -->|"Yes"| OUT["ResultSuccessSkill clears booking + urgent_message"]
```

4.5 Testing and Verification

Recommended checks:

1. Publish the project and verify `SetupFlow` runs without runtime errors.
2. Confirm custom tools are present in `project_attributes_settings_newo_tools`.
3. In `Token` mode, run availability + booking + cancellation smoke tests.
4. In `OAuth` mode, verify `pipedrive_oauth_code` leads to connector creation and business requests go through `pipedrive_connector`.
5. Verify a booking creates:

- person lookup/create in `v2`
- activity create in `v2`
- `participants` payload instead of `person_id`
- UTC `due_date` / `due_time` while user-facing confirmation remains local

Recent live evidence in this repo:

- slot truncation fix validated by a prod e2e run
- booking create validated on prod after the `participants` fix
- post-fix live booking payload used `participants` and `due_time=12:00` UTC for a 3 PM Europe/Istanbul user-facing slot

5. FAQ

Q: Does the integration use generic NAF booking tools?

A: No. It registers custom tools `pipedrive_check_availability`, `pipedrive_book_appointment`, and `pipedrive_cancel_appointment`, while overriding the generic booking tools to stay disabled.

Q: Are appointments stored as Pipedrive deals?

A: No. They are stored as Pipedrive **activities**.

Q: How is the contact linked to a booked activity?

A: In `v2`, the booking payload uses `participants` with a primary participant instead of writable `person_id`.

Q: What time format is sent to Pipedrive on booking?

A: The user-facing slot is handled in local business time, then converted to UTC before `due_date` / `due_time` are sent in the activity create payload.

Q: What happens if the caller already exists in Pipedrive?

A: The integration stores `pipedrive_person_id`, reuses that contact for booking, and loads future bookings into persona memory for the conversation.

Q: Can I use OAuth without a raw API token?

A: Yes. The integration now supports connector-based OAuth through `oauth_code` + `create_extension_connector`. Token mode remains available separately.